

SKILL – 8

2520090160 – Snigdha

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#define MAX 100
```

```
void expand_variable(char *token)
```

```
{
```

```
    if (token[0] == '$')
```

```
    {
```

```
        char *value = getenv(token + 1);
```

```
        if (value != NULL)
```

```
            printf("%s -> %s\n", token, value);
```

```
        else
```

```
            printf("%s -> Undefined variable\n", token);
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("%s -> Not a variable\n", token);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
char token[MAX];
```

```
printf("Enter a variable reference: ");
```

```
scanf("%99s", token);
```

```
expand_variable(token);
```

```
return 0;
```

```
}
```

```
snigdha@Snigdha:~/quotes_practical$ nano variable_expand.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o variable_expand variable_expand.c
snigdha@Snigdha:~/quotes_practical$ ./variable_expand
Enter a variable reference: $HOME
$HOME -> /home/snigdha
snigdha@Snigdha:~/quotes_practical$ ./variable_expand
Enter a variable reference: $PATH
$PATH -> /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local
/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubs
ystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Python314/Scripts:/mnt/c/Pr
ogram Files/Python314:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Progra
m Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Or
acle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt
/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program
 Files/Git/cmd:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Fil
es/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/snigd/AppData/Local/Microsoft/WindowsApps:/mnt/c
/Users/snigd/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/snigd/AppData/Roamin
g/npm:/mnt/c/Users/snigd/AppData/Local/Python/bin:/mnt/c/Users/snigd/AppData/Local/Programs
/Antigravity/bin:/snap/bin
snigdha@Snigdha:~/quotes_practical$ ./variable_expand
Enter a variable reference: $ABCXYZ
$ABCXYZ -> Undefined variable
snigdha@Snigdha:~/quotes_practical$ ./variable_expand
Enter a variable reference: hello
hello -> Not a variable
snigdha@Snigdha:~/quotes_practical$ |
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <unistd.h>
```

```
void builtin_pwd()
```

```
{  
    char cwd[1024];  
  
    if (getcwd(cwd, sizeof(cwd)) != NULL)  
        printf("%s\n", cwd);  
    else  
        perror("pwd");  
}
```

```
void builtin_cd(char *path)  
{  
    if (path == NULL)  
    {  
        printf("cd: missing argument\n");  
        return;  
    }
```

```
    if (chdir(path) != 0)  
        perror("cd");  
}
```

```
void builtin_exit()  
{  
    printf("Exiting shell...\n");  
    exit(0);  
}
```

```
void builtin_help()
```

```
{  
    printf("Available built-ins:\n");  
    printf(" pwd\n");  
    printf(" cd <directory>\n");  
    printf(" help\n");  
    printf(" exit\n");  
}
```

```
int main()  
{  
    char command[100];  
    char argument[100];  
  
    printf("Mini Shell\n");  
  
    while (1)  
    {  
        printf("myshell> ");  
        scanf("%99s", command);  
  
        if (strcmp(command, "pwd") == 0)  
        {  
            builtin_pwd();  
        }  
        else if (strcmp(command, "cd") == 0)  
        {  
            scanf("%99s", argument);  
            builtin_cd(argument);  
        }  
    }  
}
```

```

    }

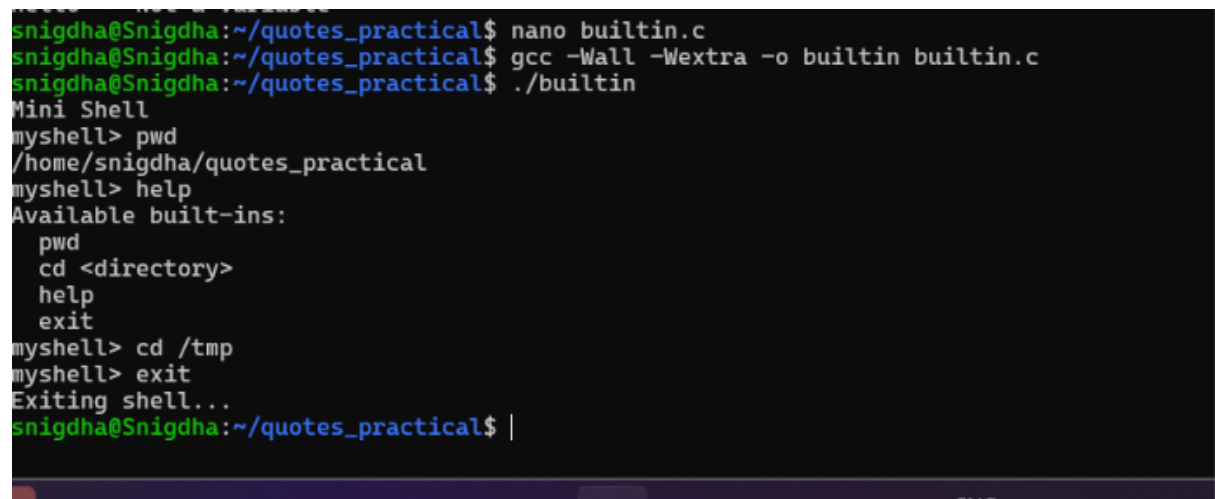
    else if (strcmp(command, "help") == 0)
    {
        builtin_help();
    }

    else if (strcmp(command, "exit") == 0)
    {
        builtin_exit();
    }

    else
    {
        printf("Invalid command: %s\n", command);
    }
}

return 0;
}

```



```

snigdha@Snigdha:~/quotes_practical$ nano builtin.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o builtin builtin.c
snigdha@Snigdha:~/quotes_practical$ ./builtin
Mini Shell
myshell> pwd
/home/snigdha/quotes_practical
myshell> help
Available built-ins:
    pwd
    cd <directory>
    help
    exit
myshell> cd /tmp
myshell> exit
Exiting shell...
snigdha@Snigdha:~/quotes_practical$ |

```